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Characterisation of cardiovascular disease (CVD) incidence and machine learning risk prediction in middle-aged and elderly populations: data from the China health and retirement longitudinal study (CHARLS)

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Abstract

Background Due to the ageing population and evolving lifestyles occurring in China, middle-aged and elderly populations have become high-risk groups for cardiovascular disease (CVD). The aim of this study was to analyse the incidence characteristics of CVD in these populations and develop a prediction model by using data from the China Health and Retirement Longitudinal Study (CHARLS).

Methods We used follow-up data from the CHARLS to analyse CVD incidence in the Chinese middle-aged and elderly population over a time span of 9 years. Five machine learning (ML) algorithms were employed for risk prediction. Data preprocessing included missing value imputation via random forest. Feature selection was performed using the Least Absolute Shrinkage and Selection Operator (Lasso CV) method with cross-validation prior to model training. The application of the synthetic minority over-sampling technique (SMOTE) to address class imbalance. Model performance was evaluated via analyses including the area under the ROC curve (AUC), precision, recall, F1 score, and SHAP plots for interpretability.

Results In accordance with the exclusion criteria, 12,580, 12,061, 11,545, and 11,619 participants were enrolled in four follow-up rounds. The cumulative incidence (CI) of CVD at 2, 4, 7, and 9 years was 2.846%, 8.971%, 17.869% and 20.518%,, respectively. Significant differences in CVD incidence were observed across gender, age, ethnicity, and region, with higher rates observed in females and in the northeast region. Ultimately, 8,080 participants and 24 features were analysed for CVD risk prediction. Five ML models were built based on these features. Although the LGB model achieves an AUC of 0.818, indicating strong overall performance, its F1 score and recall rate are relatively low, at 0.509 and 43.1%, respectively. Shapley additive explanations (SHAP) analyses revealed the importance of key

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features, such as night sleep duration, TG levels, and waist circumference, in predicting outcomes, and highlighted the nonlinear relationships between these features and CVD risk.

Conclusions Gender, age, ethnicity, and region are significant factors influencing CVD incidence. Although the LGB model demonstrates good overall performance, its low F1 score and recall rate reveal limitations in identifying high-risk cardiovascular disease patients.

Keywords Cardiovascular disease, Middle-aged and elderly individuals, Morbidity characteristics, Machine learning, Predictive modelling

Background

Cardiovascular disease (CVD) refers to a group of disorders affecting the heart and blood vessels, typically caused by atherosclerosis, thrombosis, and disruptions in blood flow, which impair the function of these organs. CVD poses a significant health threat to the Chinese population, especially among middle-aged and elderly individuals [1]. According to the China Cardiovascular Health and Disease Report 2021 [2], the number of CVD patients in China has reached 330 million, and the prevalence continues to increase. With the ageing of the population and changes in lifestyle, the middle-aged and elderly population has become a high-risk group for cardiovascular disease, especially given the presence of multiple cardiovascular risk factors, such as hypertension, diabetes mellitus, and hyperlipidaemia, which makes prediction and intervention measures for CVD particularly important [3–5]. Therefore, it is important to investigate the pathogenic factors of cardiovascular diseases in middle-aged and elderly individuals, identify high-risk groups in a timely manner, and make risk predictions [6]. The investigation of cardiovascular disease in middle-aged and elderly populations is particularly important.

CVD is characterised by a wide variety of factors that do not exist in isolation; rather, they interact in a complex network of interconnections. Although traditional CVD prediction models (e.g., the Framingham risk score and American College of Cardiology/American Heart Association (ACC/AHA) risk model) are useful to some extent in assessing risk, they typically consider only a limited number of major risk factors and tend to ignore many potentially important factors, such as the levels of inflammatory markers, psychological factors, and sleep problems. Due to the fact that most of these models are constructed based on the assumption of a linear relationship existing between variables, they may not accurately estimate the risk of individuals, especially those near the critical nodes of the nonlinear relationship. This situation calls for the development and use of more comprehensive and refined risk assessment tools.

Machine learning (ML) involves the investigation of how human learning activities can be simulated by computer programs [7]. It is an important branch of the field of artificial intelligence and has been widely used in disease prediction [8–10]. ML can effectively handle complex high-dimensional data, automatically identify nonlinear relationships and interactions between variables, and significantly improve the accuracy of disease risk prediction [11]. Compared with traditional methods, ML has demonstrated greater accuracy and efficiency in CVD prediction, which is especially suitable for multifactorial and multidimensional disease risk assessments [12, 13].

Although several machine learning-based cardiovascular disease (CVD) prediction studies exist internationally [14–18], there is still a relative lack of large-sample, long-term studies that have been conducted in middle-aged and elderly populations in China. This study aimed to resolve this research gap by analysing the incidence characteristics of CVD in the middle-aged and elderly populations and constructing a prediction model by using data from the China Health and Retirement Longitudinal Study (CHARLS) database. In this study, cardiovascular disease (CVD) refers to a broad category of diseases, including both heart diseases (e.g., coronary artery disease, heart failure) and cerebrovascular diseases (e.g., stroke). Although stroke is technically classified as a cerebrovascular disease, it is often considered within the broader context of CVD in epidemiological studies due to its shared risk factors with cardiovascular conditions. The outcome variables of interest in this study were CVD events, specifically those self-reported by participants, including heart attack, coronary artery disease, heart failure, angina, and other heart-related problems, as well as stroke. This research is expected to provide a scientific basis for the early prediction and intervention of CVD.

Methods

Data sources

The CHARLS is a nationally representative longitudinal survey program coordinated by the China National Development Research Institute (NDRI) for people aged 45 years and older, as well as their spouses. The survey assesses the social, economic and health statuses of people in the community. Detailed information on the study population has been provided in other publications [19]. The national baseline survey (wave 1) was completed in 2011–2012 and encompassed 10,257 households and

17,708 participants from 28 provinces in China. Thereafter, the participants were followed up every 2 to 3 years. This study used baseline data from 2011 and four subsequent follow-ups: the first follow-up in 2013 (wave 2), the second follow-up in 2015 (wave 3), the third follow-up in 2018 (wave 4), and the fourth follow-up in 2020 (wave 5). The CHARLS datasets can be downloaded from the CHARLS home page at http://charls.pku.edu.cn/en. The baseline and follow-up data of the CHARLS were obtained based on approval from the Peking University Ethics Review Board (IRB00001052-11015), and all of the respondents signed an informed consent form.

Subjects in the morbidity study

A total of 17,708 participants were enrolled in wave 1 and were screened to exclude subjects who (1) were younger than 45 years of age, (2) already had heart disease or stroke at the time of the baseline survey, and (3) had missing information on heart disease and stroke at the baseline survey. After screening, a total of 14,399 eligible subjects were enrolled in the baseline study. In each subsequent follow-up wave, those individuals who were lost to follow-up, or those with missing information on heart disease and stroke were excluded. Additionally, wave 2 and wave 5 provided complete and reliable records of mortality exit, so participants from these waves were included in the analysis. However, since no mortality exit data were available for wave 3 and wave 4, participants who had died during these waves were excluded. Specifically, 12,580, 12,061, 11,545, and 11,619 study subjects were enrolled in wave 2, wave 3, wave 4, and wave 5 follow-ups, respectively, and these data were used to examine CVD incidence at different follow-up time points.

Subjects in the risk prediction study

Of the 14,399 study participants who were initially screened, those individuals who lacked blood test data and those with a 20% or greater rate of missing data were excluded from the study, thus resulting in a total of 9,090 participants who met the study criteria. In a subsequent exclusion, study participants who were lost to follow-up, or had missing data related to heart disease and stroke were again excluded. A total of 8,080 study participants were ultimately analysed for CVD risk prediction over a 9-year time period.

The selection of subjects for this study and the detailed process of the study design for CVD incidence and risk prediction are shown in Fig. 1.

Definition of CVD

The outcome variables of interest in this study were cardiovascular disease (CVD) events, including both heart diseases (e.g., coronary heart disease, heart failure) and cerebrovascular diseases (e.g., stroke). Consistent with previous studies [20–22], CVD events were assessed based on the following questions: "Has your doctor ever diagnosed you with heart disease, angina, coronary heart disease, heart failure, or other heart problems?" and "Has a doctor ever diagnosed you with a stroke?". Participants with self-reported heart disease or stroke were defined as having cardiovascular disease [20, 21].

Candidate variables for the risk prediction study

Using relevant literature and information from the CHARLS dataset, this study identified a series of key candidate variables. These variables were categorized into three main groups: demographic background, health status and functioning, and blood test indicators. The demographic background data included age, gender, ethnicity, education, rural, marriage and region data. Health status and functioning included health status, lifestyle and health behaviours, functional limitations and helpers, cognition and depression and clinical examinations (height, weight, waist, systolic blood pressure, et al.). Blood test indicators included C-reactive protein (CRP), glycated haemoglobin (HbA1c), fasting glucose (FG), total cholesterol (TC), high-density lipoprotein (HDL), low-density lipoprotein (LDL), triglyceride (TG), haemoglobin (Hb), cystatin C, creatinine (Cr), urea nitrogen (UN), and uric acid (UA) levels. Moreover, this study defined the diagnostic criteria for the two diseases as follows. Hypertension was determined by fulfilling any of the following conditions: self-reported hypertension; physical examination revealing a diastolic blood pressure of no less than 90 mmHg or a systolic blood pressure of no less than 140 mmHg; and diabetes mellitus being defined as a self-reported diagnosis, physician-confirmed diagnosis or blood test revealing a fasting blood glucose level of no less than 126 mg/dl, or a glycated haemoglobin level of no less than 6.5%. Details of the definitions and measurements of the relevant variables can be found in the China Health and Ageing Report [21].

In addition, we calculated several composite indicators, including body mass index (BMI), metabolic syndrome (Mets), circadian syndrome (Circs), low handgrip strength, and muscle mass, among other indicators. Body mass index (BMI) was defined as weight (unit: kg) divided by the square of the height (unit: m). Skeletal muscle mass (ASM) for the Chinese population was estimated via a physical measurement formula reported in a previous study [23]: 0.193 * weight + 0.107 * height * 100–4.157 * gender – 0.037 * age – 2.631. If the participant was male, the gender was set to 1; otherwise, it was set to 0. Muscle mass = ASM / (height * height). Low handgrip strength was measured by using a grip strength metre and is usually calculated as being less than 28 kg for males and less than 18 kg for females [24]. Metabolic syndrome is a group of metabolism-related risk factors

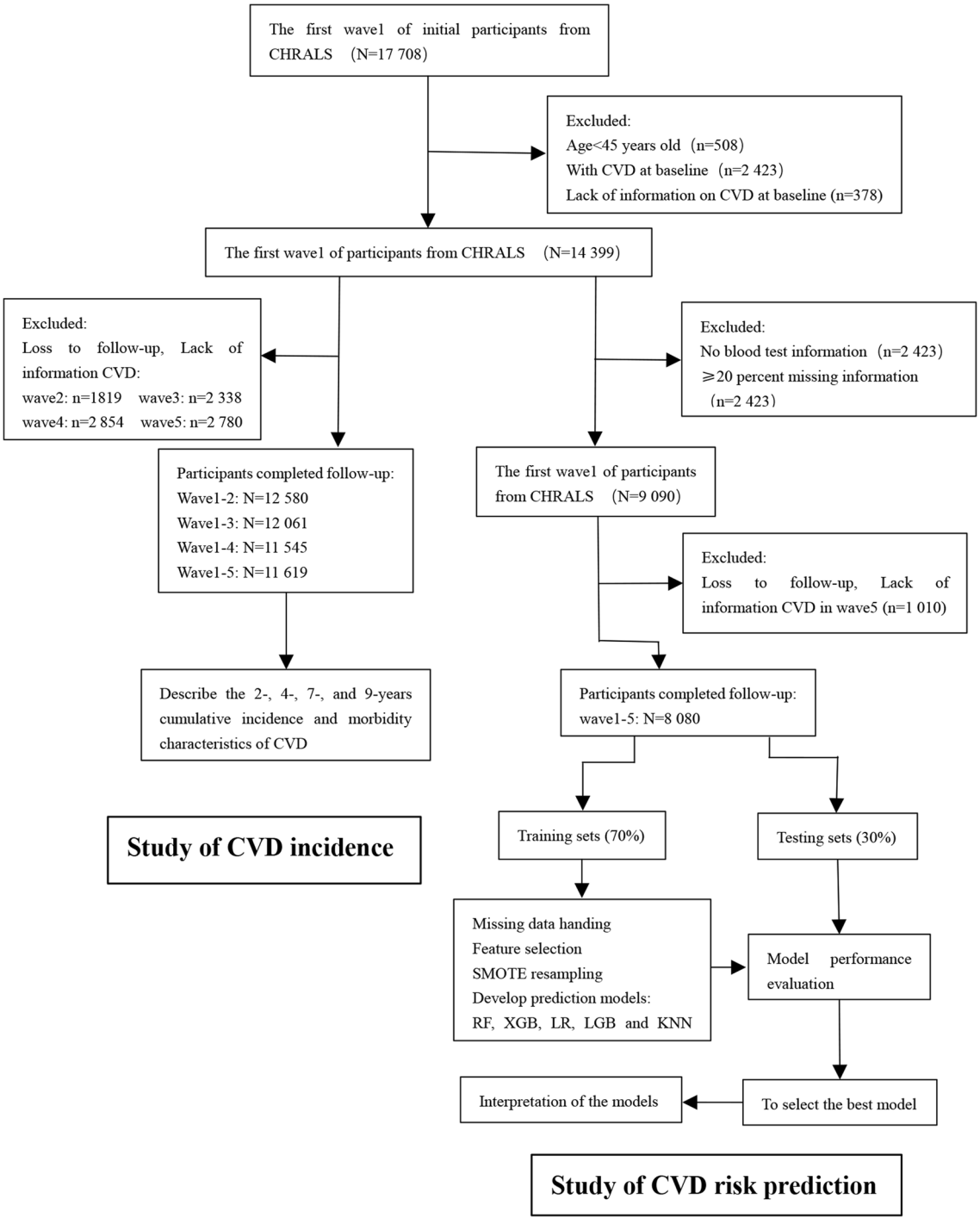


Fig. 1 The flow chart for this study

that increase the risk of cardiovascular disease and diabetes, and its diagnostic criteria are based on guidelines from the International Diabetes Federation (IDF) and the American Heart Association (AHA), among others [25]. The diagnosis requires that a patient be diagnosed with metabolic syndrome if he or she has abdominal obesity and meets at least two other criteria (elevated blood pressure, elevated fasting glucose, elevated lipids, and lowered HDL cholesterol levels). Circadian syndrome is defined as the presence of at least four of the following seven components: increased waist circumference, increased fasting glucose levels or use of related medications, increased blood pressure or use of antihypertensive medications, decreased HDL cholesterol levels or use of

lipid-lowering medications, increased triglyceride levels or use of lipid-lowering medications, short sleep duration, and incidence of depressive symptoms [26].

The study included 93 variables. For more information on the predictor variables, see Table S1.

Machine learning modelling and evaluation for risk prediction studies

In this study, we stratified the data based on cardiovascular disease (CVD) outcomes and randomly split the dataset into training and testing sets with a 7:3 ratio. The training set was used for model parameter calculation and training, while the testing set was used for evaluating model performance. For handling missing values, we employed the Random Forest Imputation method, which is a technique within Multiple Imputation. Compared to traditional imputation methods (such as mean imputation or regression imputation), Random Forest Imputation can automatically handle different types of missing data and effectively capture nonlinear relationships among features [27]. Given the diverse variables and potential nonlinearities in our dataset, Random Forest Imputation demonstrated high adaptability and predictive accuracy, while effectively avoiding biases that might arise from simpler imputation methods, thus improving data integrity and model performance.

Before model construction, feature selection was a crucial step. It not only helped prevent overfitting but also improved model training efficiency by removing redundant or irrelevant features [28]. The feature selection process was strictly confined to the training set to avoid any interaction with the test set, reducing the risk of data leakage. To achieve effective feature selection, we applied the Least Absolute Shrinkage and Selection Operator (Lasso CV) method with cross-validation on the training set. This method selects variables with non-zero coefficients, ensuring their inclusion in subsequent analysis. Given the significant class imbalance in the dataset, we further applied the Synthetic Minority Over-sampling Technique (SMOTE) to balance the class distribution (adjusting it to a 1:1 ratio) in training set. SMOTE generates synthetic samples of the minority class, optimizing model performance, and is widely used for handling imbalanced datasets [29].

After data preprocessing, we applied five classification algorithms to the training set, including k-Nearest Neighbors (kNN), Logistic Regression (LR), Random Forest Classifier (RF), Extreme Gradient Boosting (XGB), and LightGBM (LGB). kNN is one of the simplest and most commonly used supervised learning algorithms, suitable for both regression and classification tasks [30]. LR is a classical statistical method with proven effectiveness across various fields [31]. RF, by integrating multiple decision trees and using different subsets of data, reduces variance and effectively prevents overfitting, making it well-suited for handling complex, multidimensional health data [32, 33]. XGB, as an optimized gradient boosting algorithm, handles imbalanced data well and improves performance through regularization techniques [34, 35]. LGB is an efficient gradient boosting-based machine learning algorithm, especially suitable for large-scale datasets and high-dimensional problems [36]. We optimized the hyper-parameters of each model using 10-fold cross-validation and grid search to ensure robustness and reliability of the model. The optimal hyper-parameter settings for each model are listed in Table S2.

To evaluate model performance, we calculated classification metrics, including the Area Under the ROC Curve (AUC), Precision, Recall, and F1-score, on the testing set and conducted an in-depth analysis of model classification performance. Additionally, we performed model interpretability analysis using Shapley Additive Explanations (SHAP), revealing the impact of features on the prediction results by plotting the mean SHAP value bar chart and SHAP summary plot [37]. The mean SHAP value bar chart shows the average absolute SHAP value for each feature, reflecting its global impact on the model output. The SHAP summary plot provides a comprehensive view of the SHAP value distribution for each feature and its influence across the entire dataset.

All of the machine learning models were constructed and validated in Python 3.11.4. The feature filtering was conducted via the Python "scikit-learn" machine learning library (version 1.1.3). For the model construction process, we utilized the "XGBoost" (version 2.0.1), "lightgbm" (version 3.2.1) and "scikit-learn" (version 1.1.3) Python packages. Scikit-learn is an open-source Python library for machine learning and data mining, widely used for various data analysis and modeling tasks. It provides efficient tools for handling and analyzing data, particularly suitable for building predictive models, performing data mining tasks, and applying statistical learning methods. Scikit-learn offers a wide range of algorithms and tools that assist developers in accomplishing different types of machine learning tasks [38].

Statistical analysis

Normally distributed data are presented as the mean ± standard deviation ($\bar{x}\pm s$), whereas nonnormally distributed data are presented as the median and interquartile range (IQR). Count data are usually expressed as frequencies and percentages (%). Comparisons of count data between the groups were performed via the χ^2 test, whereas comparisons between groups of measured data were performed via the *t* test or *Mann-Whitney U* test, depending on the distribution of the data. Raw data cleaning was performed via Stata 17.0 software, whereas the statistical analysis was performed via SPSS 17.0.

Results

Baseline characteristics

The baseline data of this study included a total of 14,399 participants, including 7,167 males and 7,232 females. The mean age of the participants was 58.86 years (standard deviation: 9.713 years). Among the included groups, 10,492 individuals were Han Chinese, and 815 were of other ethnicities. There were 5,597 participants from the urban population, compared with 8,802 from the rural population. The participants originated from all regions of China, including 6,539 individuals from the eastern region, 2,287 from the central region, 4,644 from the western region, and 929 from the northeastern region. In terms of marital status, there were 11,562 married participants and 2,837 participants with other marital statuses.

Characterisation of CVD incidence

A total of 12,580, 12,061, 11,545, and 11,619 study participants were followed up during wave 2, wave 3, wave 4 and wave 5, respectively, and the cumulative incidences (CIs) of CVD at 2 years, 4 years, 7 years and 9 years were 2.846%, 8.971%, 17.869% and 20.518%, respectively. Details of the annual incidence rates of CVD according to gender, age group, ethnicity, place of residence, region of residence, education level, and marital status are shown in Table 1; Fig. 2. Demographic characterisation revealed that the incidence rates of CVD were higher in males than in females at 4, 7, and 9 years; the incidence rates were higher in the non-Han than in the Han ethnic groups; the incidence rates varied among regions (with the highest rate observed in Northeast China); and the CVD annual incidence rates increased with increasing age. The incidence rate of CVD also increased with age, and the abovementioned differences were statistically significant ($P<0.05$).

If participants who died were excluded from the follow-up populations in wave 2 and wave 5, the CI of CVD at 2 years and 9 years would be 2.821% (348/12,335) and 20.476% (2,253/11,003), respectively. Compared to the analysis without excluding deceased participants, the CI at 2 years and 9 years decreased by 0.025% and 0.042%, respectively.

Analysis of CVD risk prediction studies

Feature selection

In the training dataset, the Lasso CV identified 24 features with non-zero coefficients: Hypertension, Waist, CESD10, Weight, Age, Total_cognition, CRP, DBP, SBP, TG, Pulse, Glu, TC, BMI, Sleep_nap, Puff, LDL, Eyesight_distance, Sleep_night, Region, HDL, BUN, Grip and Srh, optimizing the regularization parameter λ to 0.0128.

Machine learning model construction and evaluation

Using the selected 24 features, we constructed five ML models and evaluated their AUC performance on the testing set (Fig. 3). The results show that LGB (AUC = 0.818) and RF (AUC = 0.815) exhibited similar high performance. XGB (AUC = 0.809) also performed well, while LR (AUC = 0.746) and KNN (AUC = 0.662) showed relatively poor performance. The performance metrics, including AUC, accuracy, precision, recall, and F1 score, are summarized in Table 2. In terms of accuracy, LGB (81.7%) performed the best, followed by XGB (80.0%), while KNN (66.5%) had the lowest accuracy. For precision, LGB (62.2%) led followed by XGB (55.6%), while KNN (34.2%) had the lowest precision. In recall, LR (61.42%) performed the best, while XGB had the lowest recall (44.94%). In terms of F1 score, LGB (0.509) provided the best balance, followed closely by RF (0.517), while KNN (0.427) had the lowest F1 score.

Model interpretation

The average SHAP value bar plot (Fig. 4A) ranks the features based on their average SHAP values, reflecting their importance in predicting CVD events. Sleep_night ranks first, followed by TG and waist, with HDL also playing an important role in CVD prediction. In contrast, features such as BUN and Puff contribute less to the model's prediction. The SHAP summary plot (Fig. 4B) illustrates the direction and magnitude of the impact of each feature on the model's prediction. Key influencing features include abnormal sleep duration, either too short or too long (blue), which are associated with negative SHAP values. In contrast, normal sleep duration (close to the red center) has a positive impact on CVD prediction. Higher triglyceride levels and larger waist circumference (red) significantly increase SHAP values, indicating a positive relationship with CVD risk. High HDL levels (red) lower SHAP values, showing a protective effect on cardiovascular health. Features with moderate influence include regional differences, Total_cognition, and Srh, which have a certain impact on CVD risk prediction but are more complex. Lower total cognition and Srh scores negatively affect cardiovascular health, while higher levels of C-reactive protein and blood glucose (red) may significantly contribute to CVD risk. BUN, Puff, and Pulse have a smaller contribution to CVD prediction but can still explain some individual risks.

Discussion

In this study, we analysed the characteristics of the middle-aged and elderly population in China in terms of CVD incidence via the CHARLS data. The results of the study revealed that the cumulative incidence of CVD tended to increase between 2011 and 2020, with specific incidence rates of 2.846%, 8.971%, 17.869% and

Table 1 Incidence characteristics of cardiovascular diseases

variables	wave1-2			wave1-3			wave1-4			wave1-5		
	n	CVD	CI (%)	*n*	CVD	CI (%)	*n*	CVD	CI (%)	*n*	CVD	CI (%)
Gender												
Male	6 230	159	2.552	5 953	476	7.996	5 625	903	16.053	5 660	1 063	18.781
Female	6 350	199	3.134	6 108	606	9.921	5 920	1 160	19.595	5 959	1 321	22.168
P	0.056			< 0.001			< 0.001			< 0.001		
Age(year)												
45 ≤ age<55	4 493	98	2.181	4 432	292	6.588	4 387	582	13.266	4 484	697	15.544
55 ≤ age<65	4 778	135	2.825	4 637	423	9.122	4 544	888	19.542	4 573	1 051	22.983
65 ≤ age<75	2 294	86	3.749	2 137	263	12.307	1 946	469	24.101	1 901	522	27.459
≥ 75	1 015	39	3.842	855	104	12.164	668	124	18.563	661	114	17.247
P	0.001			< 0.001			< 0.001			< 0.001		
Ethnicity												
Han	9 609	243	2.529	9 712	819	8.433	10 486	1 861	17.747	9 870	2 069	20.963
Others	743	23	3.096	742	88	11.860	814	192	23.587	715	174	24.336
P	0.412			0.002			< 0.001			0.037		
Rural												
City	4 579	135	2.948	4 262	419	9.831	4 094	784	19.150	4 105	858	20.901
Country	8 001	223	2.787	7 799	663	8.501	7 451	1 279	17.165	7 514	1 526	20.309
P	0.64			0.016			0.008			0.464		
Region												
North-eastern	799	32	4.005	744	116	15.591	688	206	29.942	619	190	30.695
Eastern	5 666	148	2.612	5 502	443	8.052	5 239	871	16.625	5 347	1 027	19.207
Central	2 039	67	3.286	1 915	169	8.825	1 871	353	18.867	1 925	441	22.909
Western	4 076	111	2.723	3 900	354	9.077	3 747	633	16.894	3 728	726	19.474
P	0.084			< 0.001			< 0.001			< 0.001		
Education												
Below primary school	5 854	185	3.160	5 613	528	9.407	5 275	947	17.953	5 299	1 104	20.834
Primary school	2 645	66	2.495	2 556	216	8.451	2 450	451	18.408	2 457	527	21.449
Middle school	2 596	62	2.388	2 545	207	8.134	2 452	405	16.517	2 488	486	19.534
High school or above	1 473	44	2.987	1 334	130	9.745	1 356	258	19.027	1 362	265	19.457
P	0.148			0.15			0.187			0.256		
Marriage												
Married	10 227	278	2.718	9 810	848	8.644	9 467	1 696	17.915	9 494	1 962	20.666
Other	2 353	80	3.400	2 251	234	10.395	2 078	367	17.661	2 125	422	19.859
P	0.085			0.01			0.809			< 0.422		
Total	12 580	358	2.846	12 061	1082	8.971	11 545	2063	17.869	11 619	2384	20.518

CI: Cumulative incidence

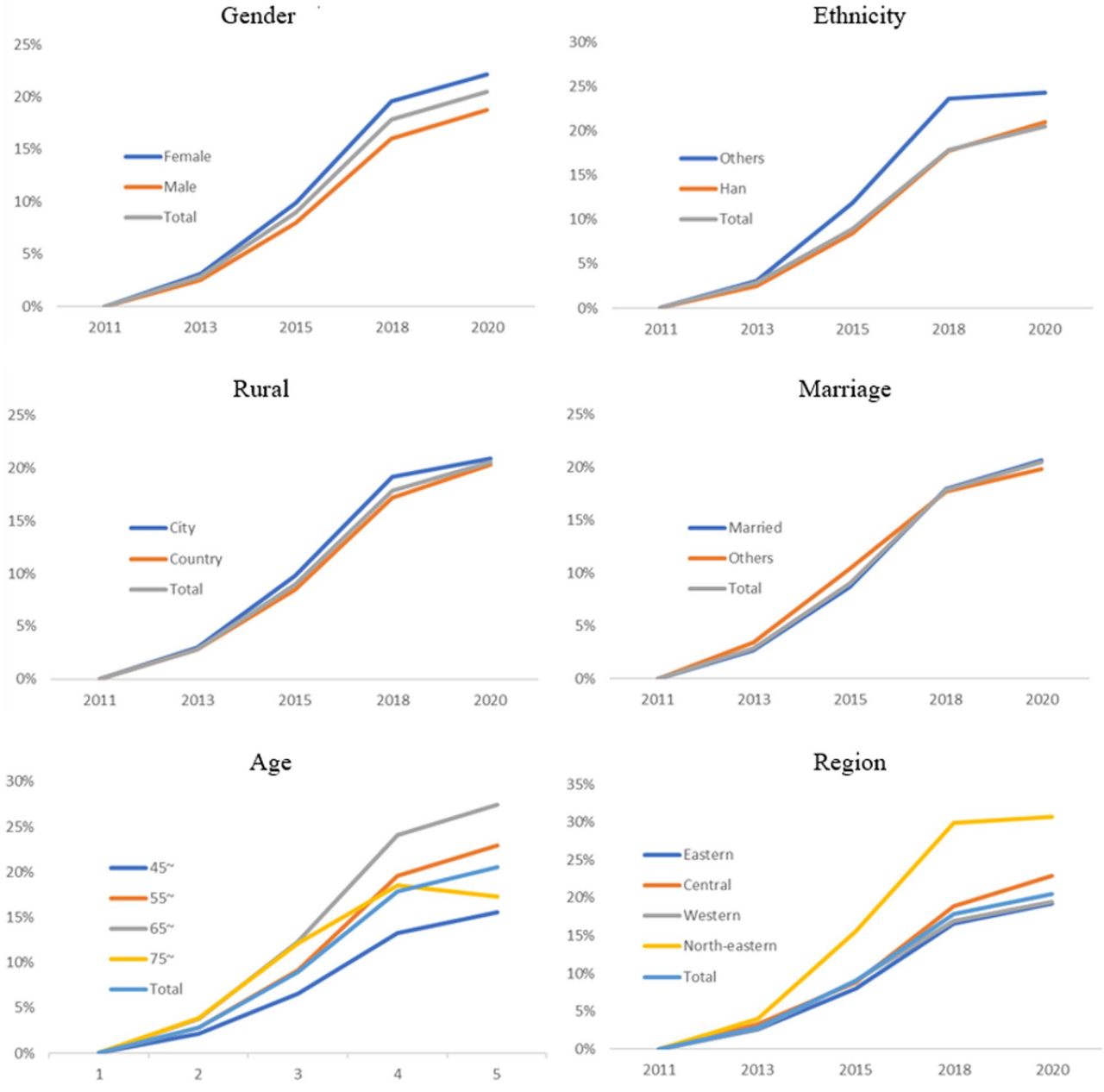


Fig. 2 Cumulative incidence and characteristics of CVD from 2011 to 2020

20.518% observed at the 2-, 4-, 7-, and 9-year observation points, respectively, which was consistent with previous research findings [39–41]. Compared with the study of Shinya Nakada et al. [42], the present study revealed a higher 4-year incidence rate. Compared with the study of Dachuan Guo [43], the 7-year incidence rate appeared to be lower, and this difference may be due to differences in population characteristics and study methods. With respect to gender, the CVD incidence rate was consistently higher in women than in men, and this difference increased over time, which is consistent with the results of previous studies [39, 44]. However, studies in the Mediterranean region have reported a greater prevalence of CVD in men than in women [14], which is contrary to the results of the present study. In addition, the present study confirmed that age is an important factor in the development of CVD, with a significant increase in CVD incidence being observed with increasing age. From an ethnic group perspective, the incidence of CVD in Han Chinese individuals was significantly lower than that in other ethnic groups from the 4th year of follow-up onwards. With respect to the effect of geographic residence, urban residents had higher CVD incidence rates than did rural residents during the 4- and 7-year

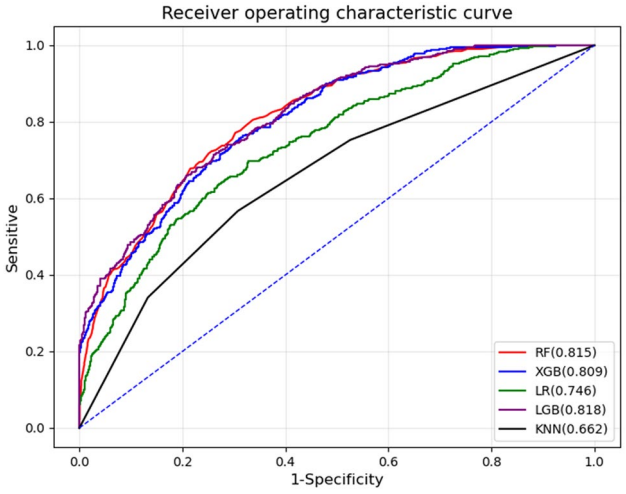


Fig. 3 Receiver operating characteristic curve (ROC) for five ML models on the testing set

Table 2 The predictive performance metrics for five ML models in testing set

Model	Accuracy(%)	Precision(%)	Recall(%)	F1-score	AUC
RF	78.8	51.9	51.5	0.517	0.815
XGB	80.0	55.6	44.9	0.497	0.809
LR	71.6	40.4	61.4	0.488	0.746
LGB	81.7	62.2	43.1	0.509	0.818
KNN	66.5	34.2	56.7	0.427	0.662

follow-up periods, whereas the differences were not significant during the 2- and 9-year follow-up periods. Regional differences were also significant, with a significantly greater incidence of CVD observed in Northeast China than in other regions during the follow-up period of four years and beyond. This could be attributed to the region's high-salt diet and cold climate [45]. These differences in gender, age, ethnicity, and place of residence provide important perspectives for CVD incidence prediction. However, traditional prediction models have mostly relied on statistical methods such as linear regression, which limits their efficacy in addressing complex feature interactions and high-dimensional data imbalance problems.

To improve the accuracy of CVD incidence risk prediction, ML methods were used in this study. ML has advantages in addressing nonlinear relationships, complex feature interactions, and high-dimensional data and data imbalance problems, thus providing a more powerful tool for CVD risk prediction. It has been shown that ML methods can effectively identify and extract key features, thus improving the accuracy and usefulness of predictions [14–18]. For example, Oh et al. developed four ML CVD classifiers based on Korean data and identified key risk factors via SHAP analysis [17]. Zhuang et al. applied randomized survival forests to analyse US community

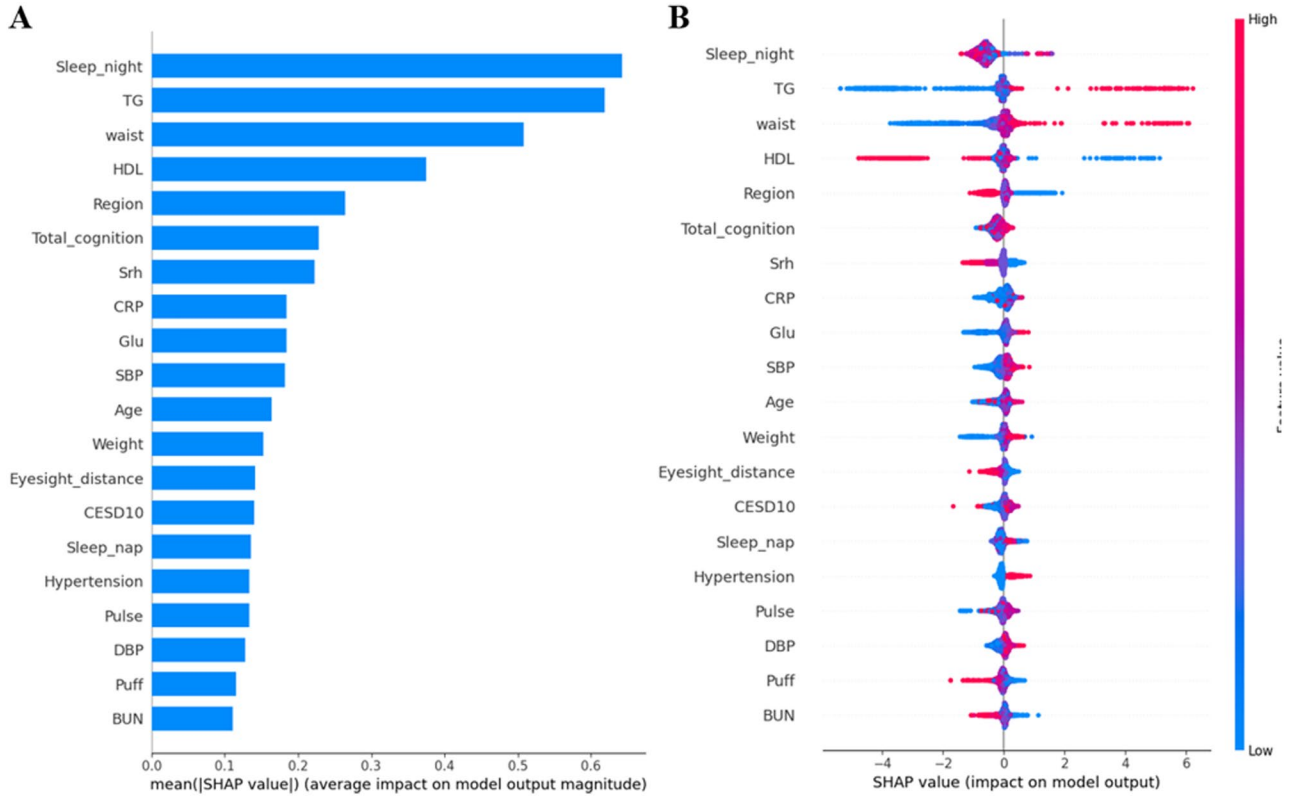


Fig. 4 The mean SHAP value bar chart (**A**) and the SHAP summary plot (**B**) based on LGB algorithmic models

atherosclerosis data to explore the process by which subclinical diseases progress to CVD events [16]. You et al. constructed a 10-year CVD incidence risk prediction model based on UK biobank data and demonstrated superior performance to traditional clinical models [15]. These studies suggest that ML algorithms have significant advantages in CVD incidence prediction; however, large-sample, long-term prediction studies based on Chinese middle-aged and elderly populations are lacking.

In this study, we developed five ML models to predict the 9-year CVD incidence among individuals aged 45 and above in China, using baseline data from the 2011 CHARLS database and follow-up data from 2020. Despite an AUC of 0.818, indicating strong overall performance, the LGB model shows limitations in identifying high-risk CVD patients, as evidenced by its F1 score of 0.509 and recall rate of 43.1%. Similar issues are observed in other models (e.g., RF, XGB, and LR), with generally low recall and F1 scores, particularly in the KNN model, which has a recall rate of 56.7% but an F1 score of only 0.427. These results suggest that while the models exhibit good overall classification performance, they still struggle to identify high-risk patients effectively. Several factors may contribute to the low sensitivity and F1 scores. First, CVD is a relatively rare event in the dataset (with fewer positive samples), which may lead to biased model predictions, particularly for high-risk patients. While the model can distinguish most negative samples accurately, its performance on identifying positive cases remains suboptimal. Second, despite applying SMOTE oversampling and feature selection, some critical predictors of high-risk patients may not be adequately captured, limiting the model's ability to identify those at greatest risk. Lastly, the current decision threshold may not be optimized for maximizing recall, resulting in the model being biased towards predicting negative cases, thus reducing its sensitivity to high-risk patients. In future research, we will explore additional balancing methods, optimize feature selection, and adjust decision thresholds to further improve the model's performance, particularly in identifying high-risk patients.

The decision tree structure of LGB is relatively simple, making it conducive to explainability analysis using methods such as SHAP, which helps to analyze the importance of features and their impact on model predictions [36, 37]. Therefore, we applied SHAP to the LGB model for interpretability analysis. By combining the average SHAP value bar chart and SHAP summary plot, we identified the most important features, including night sleep duration, TG levels, and waist circumference. These features are closely related to lifestyle, metabolic syndrome, and lipid metabolism health, suggesting that interventions targeting these high-impact features could help reduce CVD risk. Furthermore, the relationships between these features and CVD events are complex, warranting further exploration of their nonlinear relationships with CVD risk, which holds significant theoretical value for future personalized intervention strategies. Other features, such as Total_cognition, Srh, and CRP, although having relatively smaller average impacts, also contribute to CVD event prediction to some extent. These features may be associated with CVD through various physiological, psychological, or environmental mechanisms. Future research and clinical practice should focus on deeper analysis and interventions targeting these key features to better prevent and manage CVD events. Additionally, attention should be given to potential interactions between features and the moderating effects of individual differences on these features' influence.

Although this study provides valuable insights into CVD prediction in the Chinese middle-aged and elderly populations, it has several limitations. Firstly, the study was solely based on CHARLS data, and the external applicability of the results may be limited. Secondly, although longitudinal data were used for the analysis, the effects of confounding factors may still be present, as causality could not be clarified. Moreover, no external validation was conducted in this study, which may affect the generalizability and reliability of the model. Additionally, the study relied on self-reported data to define CVD events, which is subject to recall bias and underreporting, particularly for undiagnosed conditions. Despite efforts to validate the self-reported diagnoses, the lack of clinical confirmation (e.g., ECG, echocardiogram) may lead to some underestimation of CVD cases. Finally, after excluding deceased participants, the CI of CVD at 2 years and 9 years was 0.025% and 0.042% higher, respectively. Therefore, the CI at 4 years and 7 years may have been underestimated by approximately 0.025–0.042%, potentially introducing survival bias and affecting the accuracy of the results.

In light of these limitations, future studies should expand the sample sources, consider additional risk factors, conduct external validation, and incorporate clinical diagnostic data to enhance the applicability, reliability, and stability of the model.

Conclusions

This study analysed the incidence of cardiovascular disease (CVD) and its risk factors in middle-aged and elderly populations in China based on the CHARLS data. The results revealed that the cumulative incidence of CVD is increasing, and there are significant differences in gender, age, ethnicity, and region, with particularly high rates observed among women and in Northeast China. Although the LGB model achieves an AUC of 0.818, indicating good overall performance, its F1 score and recall rate are only 0.509 and 43.1%, revealing its limitations

in identifying high-risk cardiovascular disease patients. SHAP analyses revealed the importance of key features, such as night sleep duration, TG levels, and waist circumference, in predicting outcomes, and highlighted the nonlinear relationships between these features and CVD risk. Although this study provides a new perspective for CVD risk prediction, there are still limitations, such as the external applicability of the sample and the omission of potential factors. Furthermore, there is a need to expand the sample size and validate the generalizability of the model in the future.

Abbreviations

CVD	Cardiovascular disease
CHARLS	China Health and Retirement Longitudinal Study
CI	Cumulative incidence
AHA	American heart association
ACC	American college of cardiology
ML	Machine learning
NDRI	China national development research institute
ADL	Activities of daily living scale
IADL	Instrumental daily living scale
CRP	C-reactive protein
HbA1c	Glycated hemoglobin
FG	Fasting glucose
TC	Total cholesterol
HDL	High density lipoprotein
LDL	Low density lipoprotein
TG	Triglycerides
Hb	Hemoglobin
Cr	Creatinine
UN	Urea nitrogen
UA	Uric acid
BMI	Body mass index
Mets	Metabolic syndrome
Circs	Circadian syndrome
ASM	Skeletal muscle mass
IDF	International diabetes federation
SMOTE	Synthetic minority over-sampling technique
KMO	Kaiser-Meyer-Olkin
PCA	Principal component analysis
SHAP	Shapley additive explanations
KNN	k-nearest neighbor
LR	Logistic regression
RF	Random forest
XGB	XGBoost
LGB	LightGBM
IQR	Interquartile range
SBP	Systolic blood pressure
DBP	Diastolic blood pressure
AUC	Area under the ROC curve

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

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Author contributions

Qing Huang and Zihao Jiang: formal analysis, data curation, writing—original draft, and visualization. Bo Shi and Jiaxu Meng: machine learning modelling and analysis. Li Shu and Fuyong Hu: supervision, project administration, writing—review, and editing. Jing Mi: conceptualization, writing—review, editing, supervision, and funding acquisition.

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Data availability

Data is provided within the supplementary information files.

Declarations

Ethics approval

The baseline and follow-up surveys of CHARLS were obtained from the Peking University Ethics Review Board (IRB00001052-11015) approval, and all respondents signed an informed consent form.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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